

CSE428: Image Processing

Lecture 4: Point Processing - Part 2

June 11, 2022



Histogram

Definition

- For each gray level, **how many** pixels have that exact value

$$h_r(r_k) = n_k = |\{I(x, y) | I(x, y) = r_k\}| \quad r_k = 0, 1, 2, \dots, 255$$

n_k is the number of pixels with with intensity r_k

- Normalized histogram: for each gray level, **what percentage** of pixels have that exact value. I.e., “unnormalized” histogram divided by the total number of pixels

$$f_r(r_k) = \frac{n_k}{N_T}$$

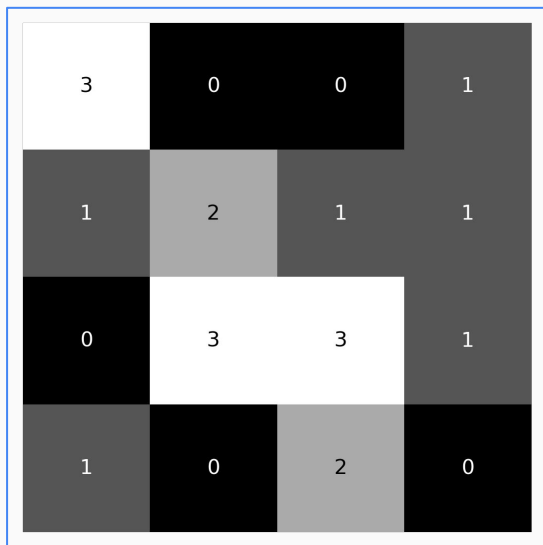
n_k is the number of pixels with with intensity r_k and N_T is the total number of pixels

- Normalized histogram is also called the probability density function (PDF) - why?

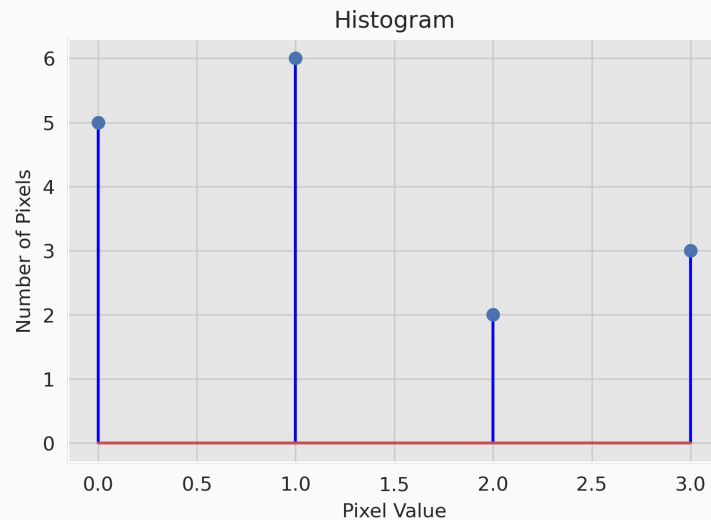
Histogram

Example

A (4 x 4) image with 2 bit or 4 quantization levels



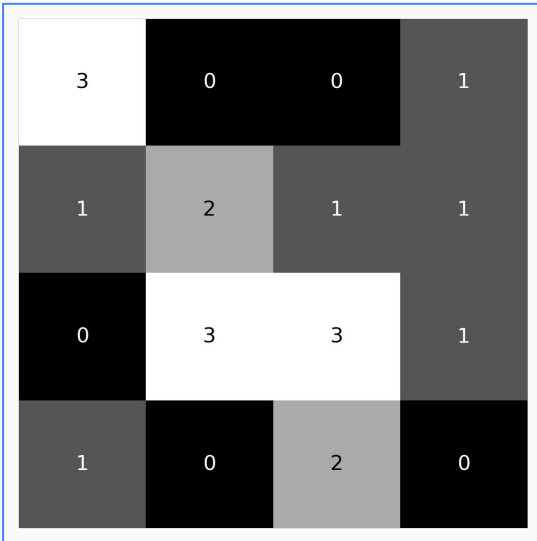
r_k	n_k
0	5
1	6
2	2
3	3



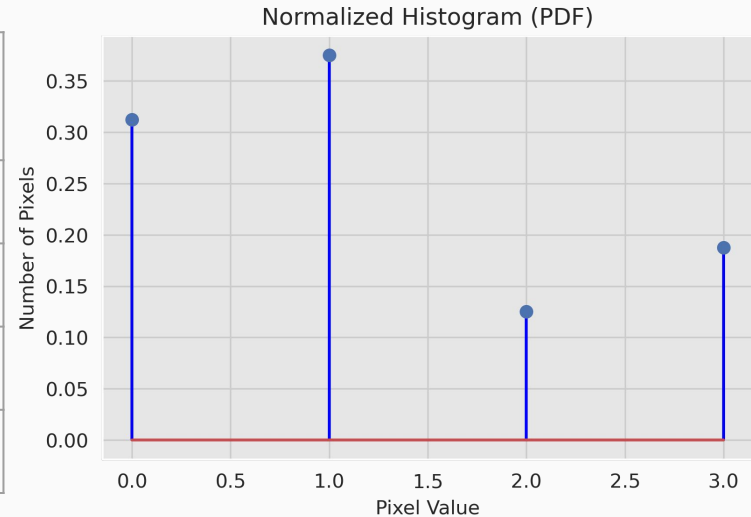
Histogram

Example

A (4 x 4) image with 2 bit or 4 quantization levels



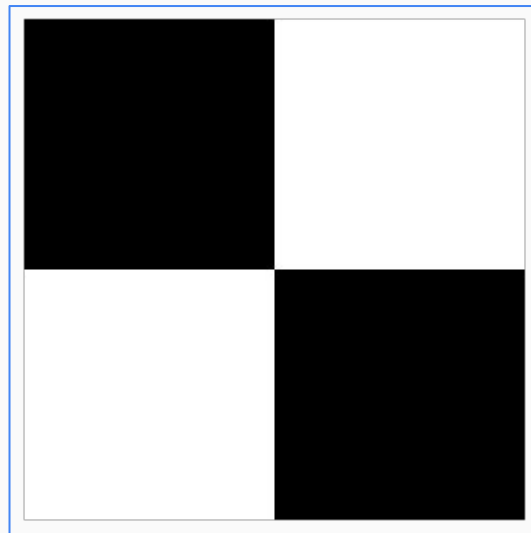
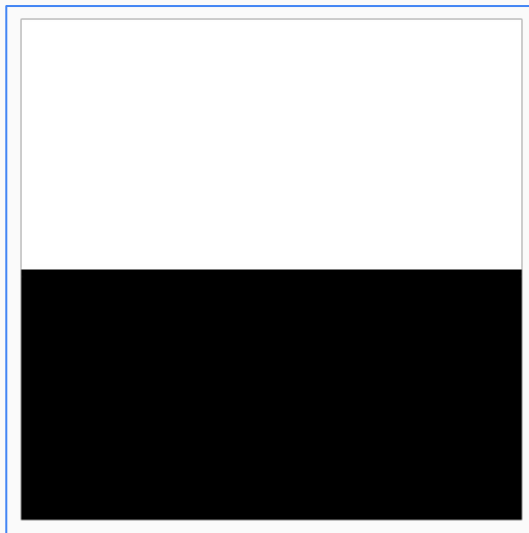
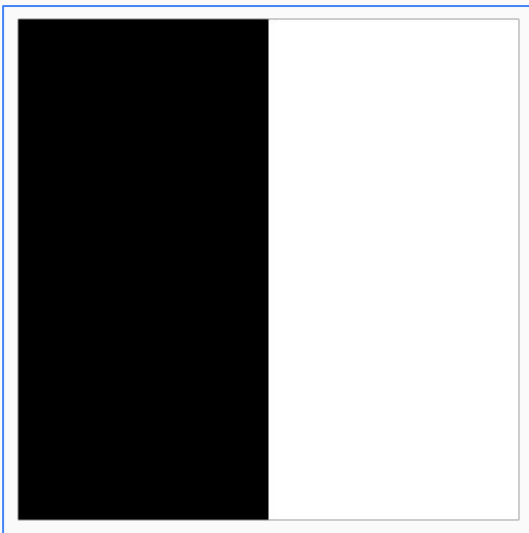
r_k	n_k	$f_r(r_k)$
0	5	0.31
1	6	0.38
2	2	0.12
3	3	0.19



Histogram

Histograms are not unique!

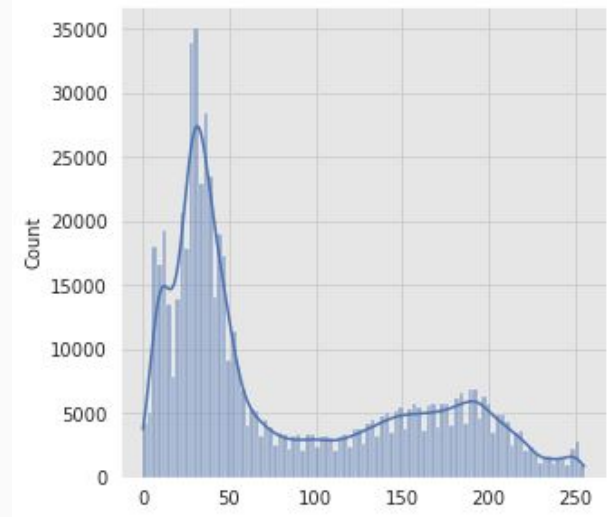
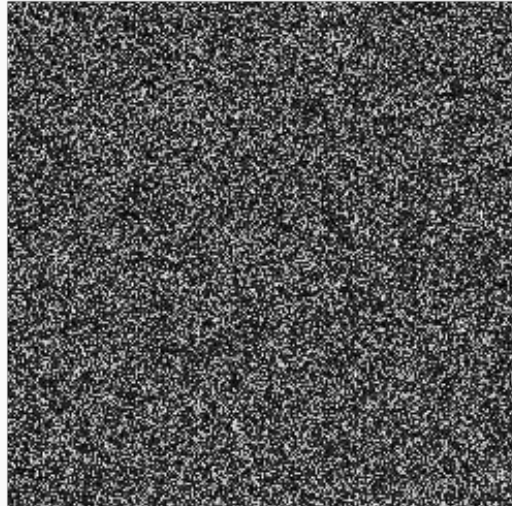
All of these images have the exact same histogram (equal number of black and white)



Histogram

Histograms are not unique!

The second image is a pixel permutation of the first image, hence same histogram



Cumulative Histogram

Definition

- For a given gray level, **how many** pixels have values $\leq$ that value.

$$H_r(r) = n_k = |\{I(x, y) | I(x, y) \leq r\}| = \int_{-\infty}^r h_r(x) dx \quad r \in \mathbb{R}$$

n_k is the number of pixels with with intensity $\leq r$.

- Integration of histogram = cumulative histogram.
- Normalized cumulative histogram: for each gray level, **what percentage** of pixels have values $\leq$ that value. Also now as cumulative density function (CDF).

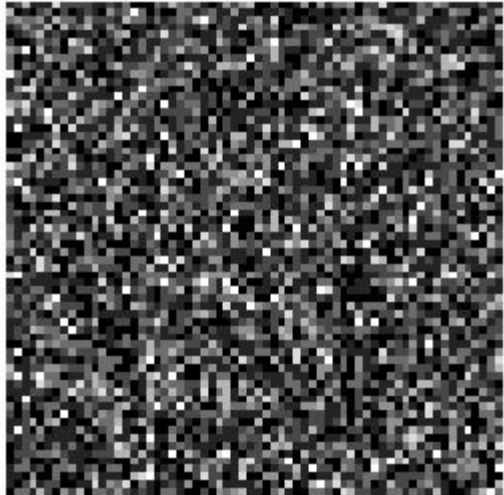
$$F_r(r) = \frac{n_k}{N_T} = \int_{-\infty}^r f_r(x) dx$$

- Interpretation of CDF(r) = probability that a randomly chosen pixel will have value $\leq r$

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

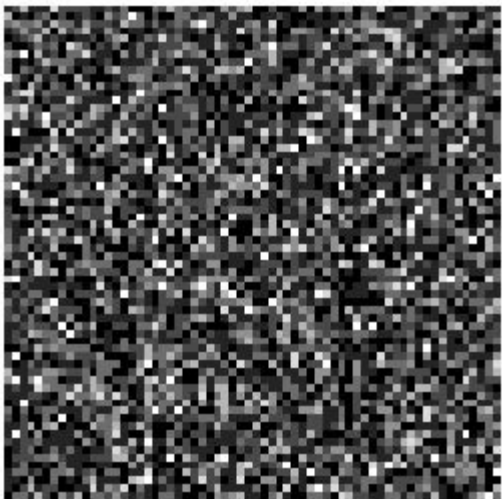


r_k	$h(r_k)$
0	790
1	1023
2	850
3	656
4	329
5	245
6	122
7	81

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

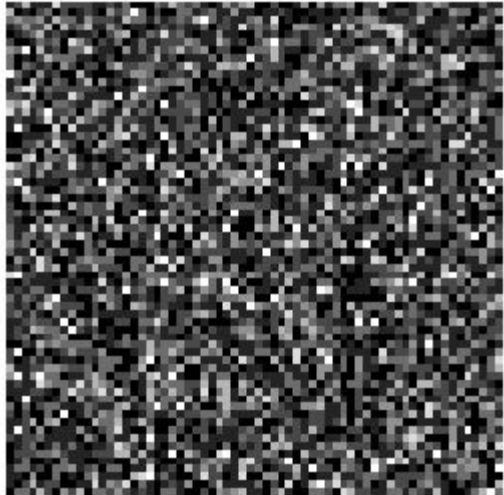


r_k	$h(r_k)$	$f(r_k)$ [PDF]
0	790	0.19 (=790/4096)
1	1023	0.25 (=1023/4096)
2	850	
3	656	
4	329	
5	245	
6	122	
7	81	

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

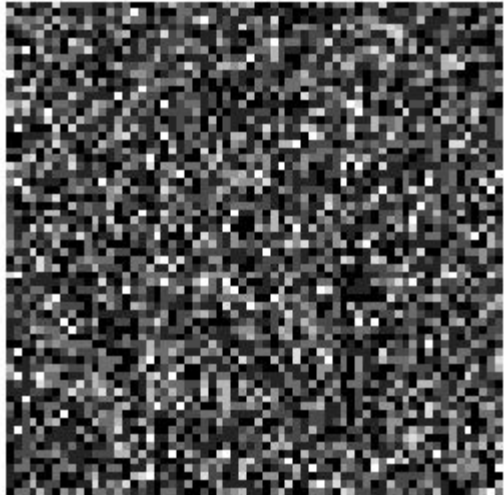


r_k	$h(r_k)$	$f(r_k)$ [PDF]
0	790	0.19
1	1023	0.25
2	850	0.21
3	656	0.16
4	329	0.08
5	245	0.06
6	122	0.03
7	81	0.02

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

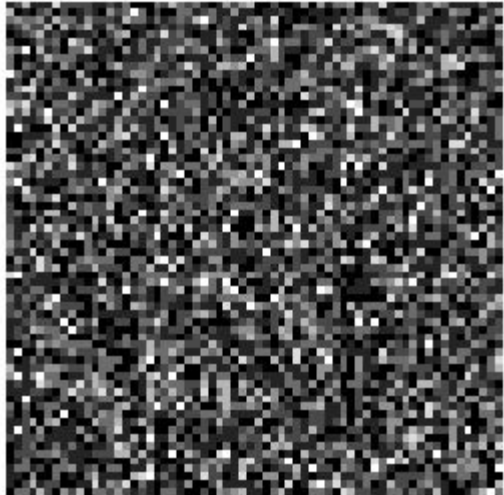


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$
0	790	0.19	790
1	1023	0.25	
2	850	0.21	
3	656	0.16	
4	329	0.08	
5	245	0.06	
6	122	0.03	
7	81	0.02	

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

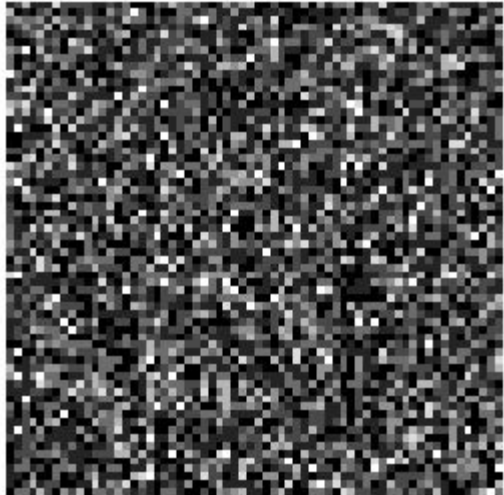


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$
0	790	0.19	790
1	1023	0.25	1813 (=790+1023)
2	850	0.21	
3	656	0.16	
4	329	0.08	
5	245	0.06	
6	122	0.03	
7	81	0.02	

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

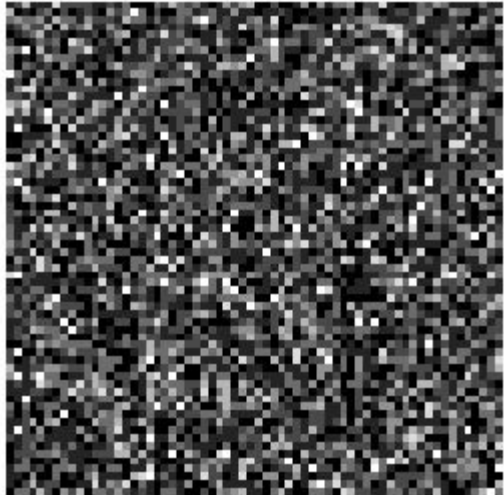


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$
0	790	0.19	790
1	1023	0.25	1813
2	850	0.21	2663 = (1813+850)
3	656	0.16	
4	329	0.08	
5	245	0.06	
6	122	0.03	
7	81	0.02	

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

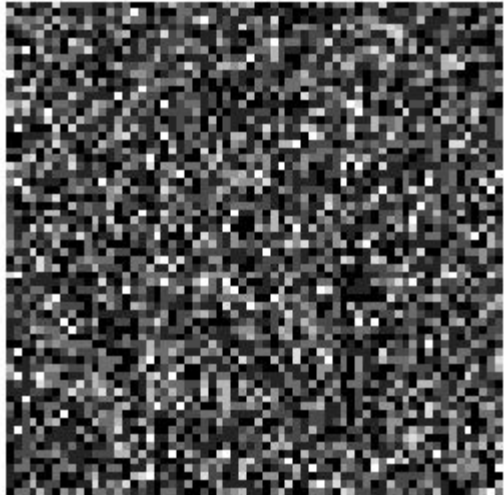


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$
0	790	0.19	790
1	1023	0.25	1813
2	850	0.21	2663
3	656	0.16	3319
4	329	0.08	3648
5	245	0.06	3893
6	122	0.03	4015
7	81	0.02	4096

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

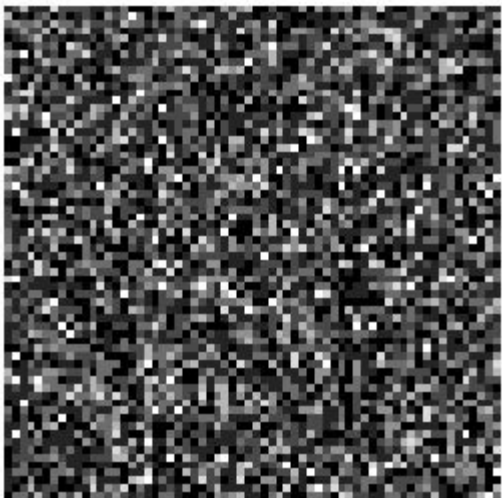


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]
0	790	0.19	790	0.19
1	1023	0.25	1813	0.44
2	850	0.21	2663	0.65
3	656	0.16	3319	0.81
4	329	0.08	3648	0.89
5	245	0.06	3893	0.95
6	122	0.03	4015	0.98
7	81	0.02	4096	1.00

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

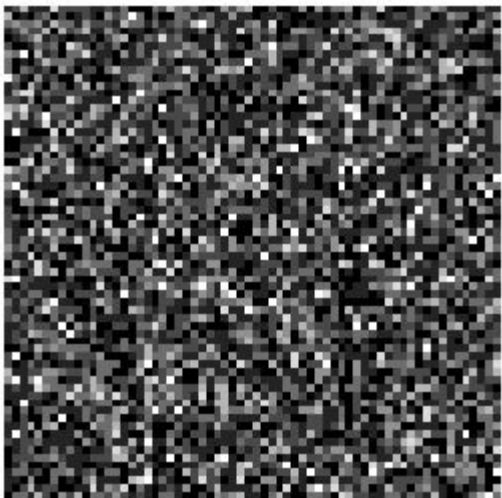


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]
0	790	0.19	790	0.19
1	1023	0.25	1813 (=790+1023)	0.44
2	850	0.21	2663	0.65
3	656	0.16	3319	0.81
4	329	0.08	3648	0.89
5	245	0.06	3893	0.95
6	122	0.03	4015	0.98
7	81	0.02	4096	1.00

Cumulative Histogram

Example

A random (64 x 64) image with 3 bit or 8 quantization levels

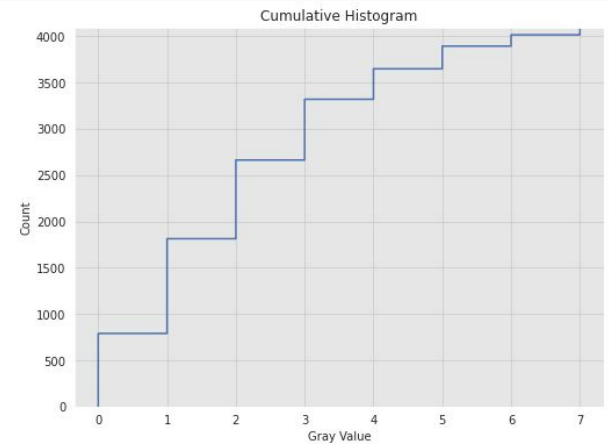
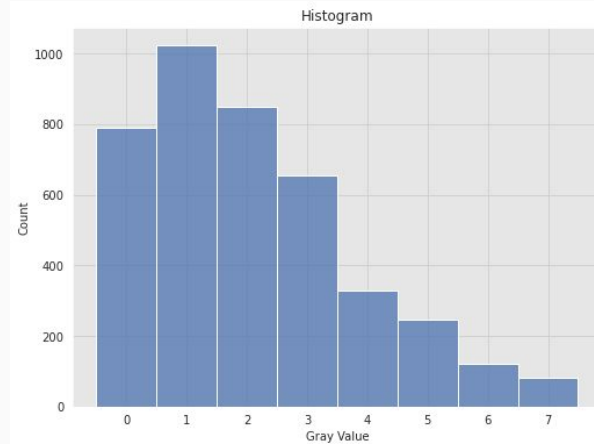
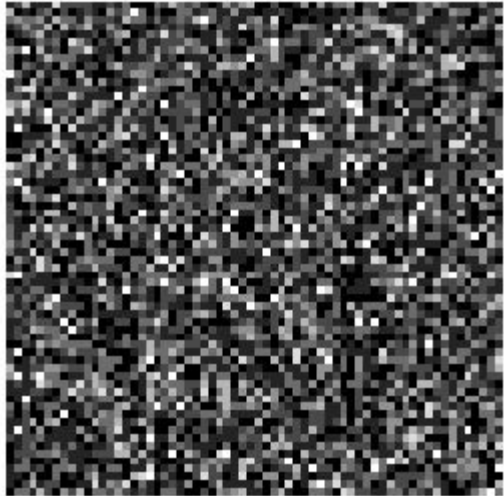


r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]
0	790	0.19	790	0.19
1	1023	0.25	1813 (=790+1023)	0.44
2	850	0.21	2663 (=1813+850)	0.65
3	656	0.16	3319	0.81
4	329	0.08	3648	0.89
5	245	0.06	3893	0.95
6	122	0.03	4015	0.98
7	81	0.02	4096	1.00

Cumulative Histogram

Example

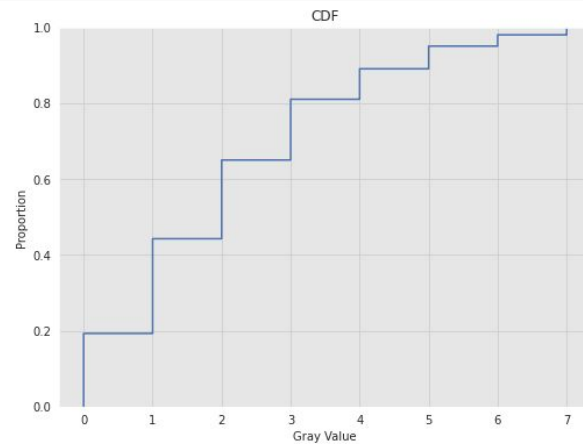
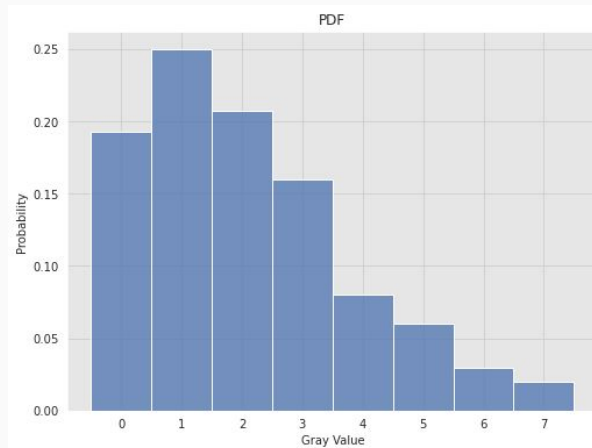
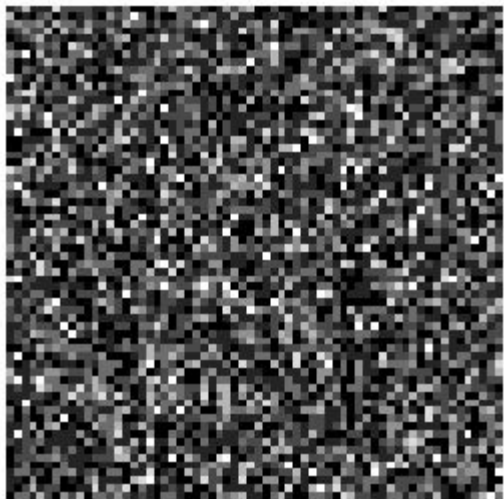
A random (64 x 64) image with 3 bit or 8 quantization levels



Cumulative Histogram

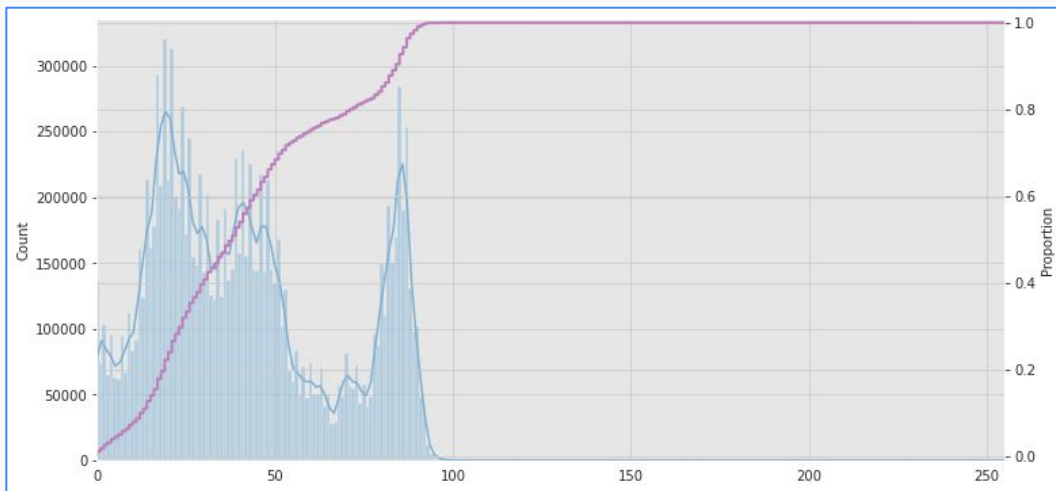
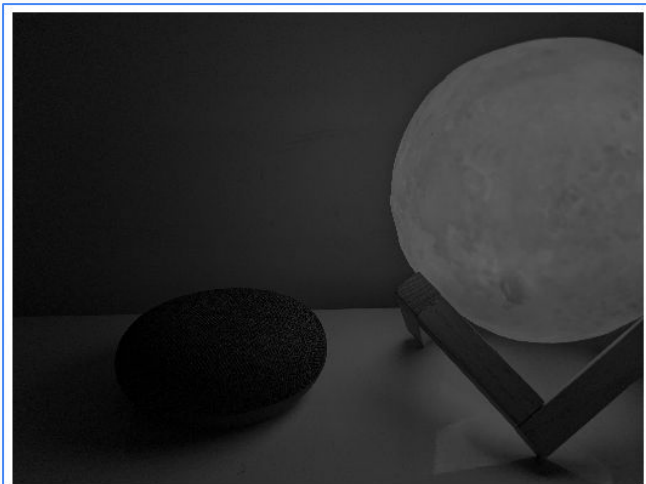
Example

A random (64 x 64) image with 3 bit or 8 quantization levels



Histogram

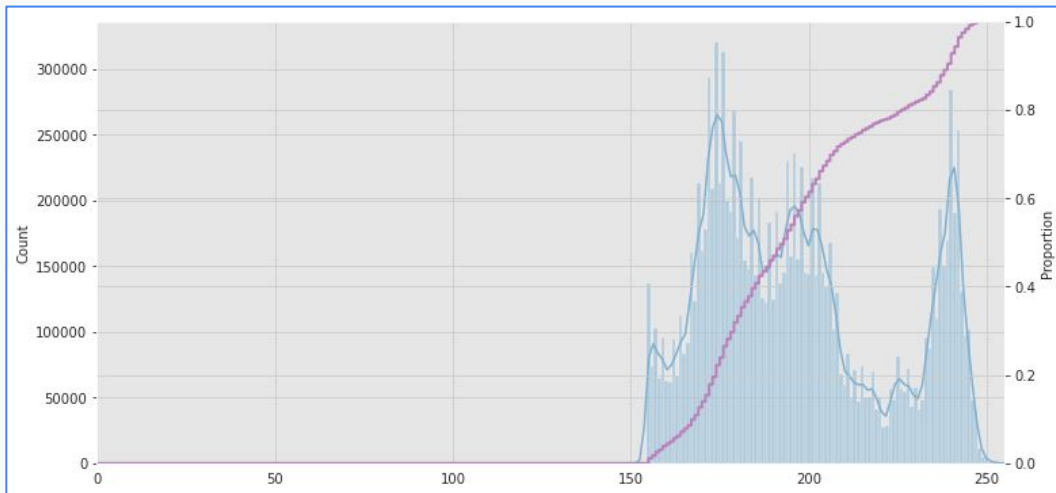
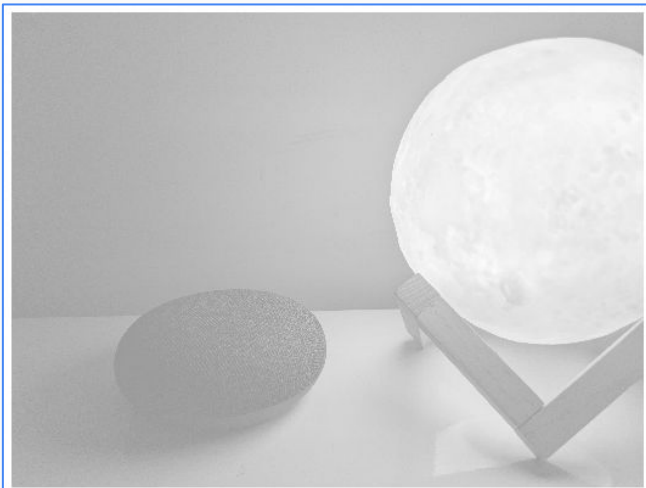
More Example



- Histogram of a dark image.
- The histogram is concentrated in the low-range gray values

Histogram

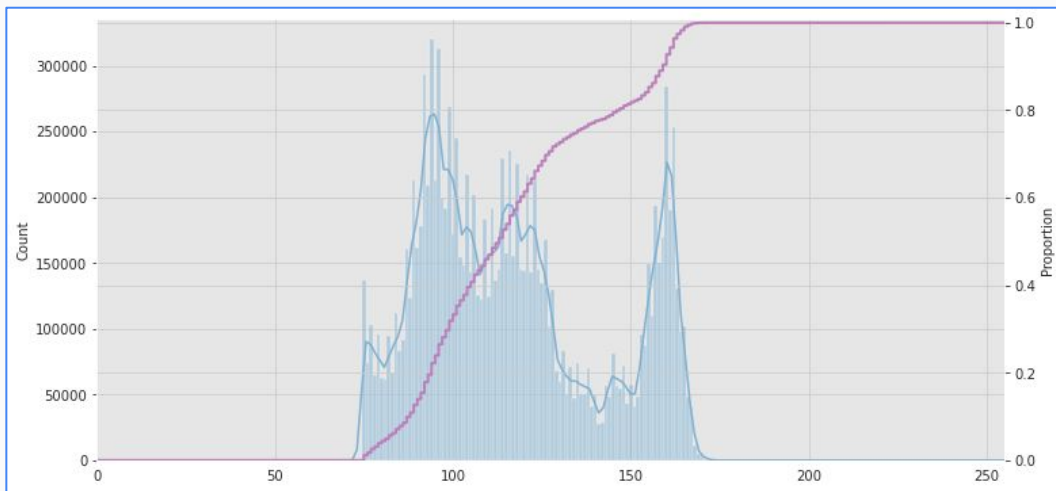
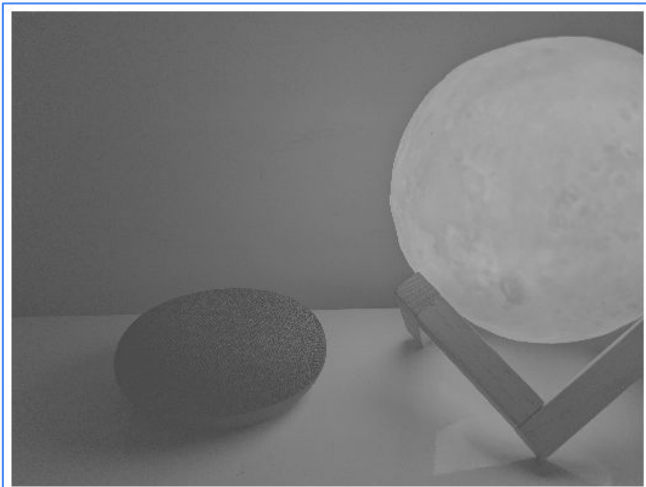
More Example



- Histogram of a bright image.
- The histogram is concentrated in the high-range gray values

Histogram

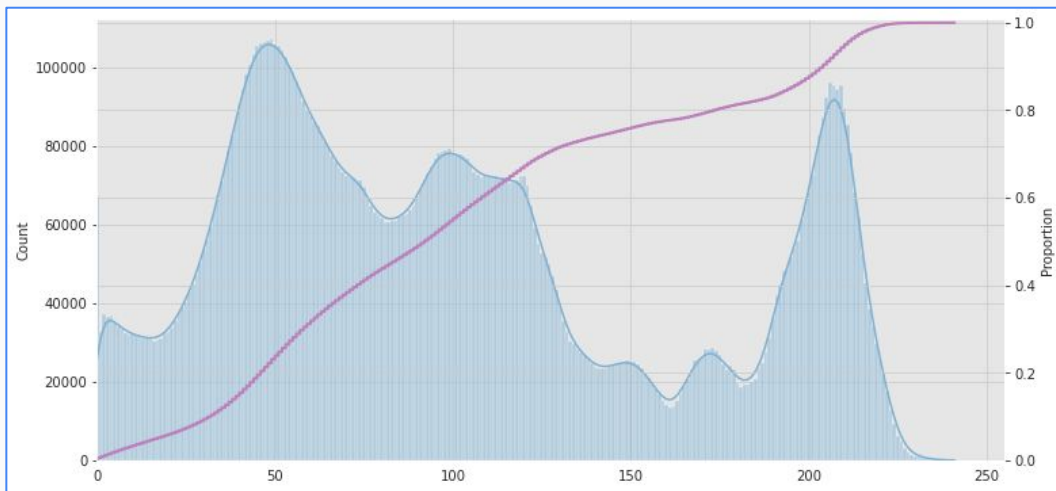
More Example



- Histogram of a low contrast image
- The histogram is concentrated in the mid-range gray values
- The image looks “washed-out”

Histogram

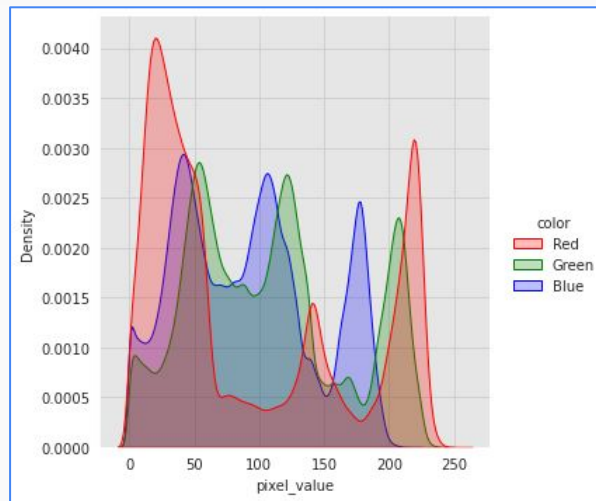
More Example



- Histogram of a high contrast image
- The histogram is (somewhat) uniformly distributed
- Contrast can be further increased by “flattening” the histogram

Histogram

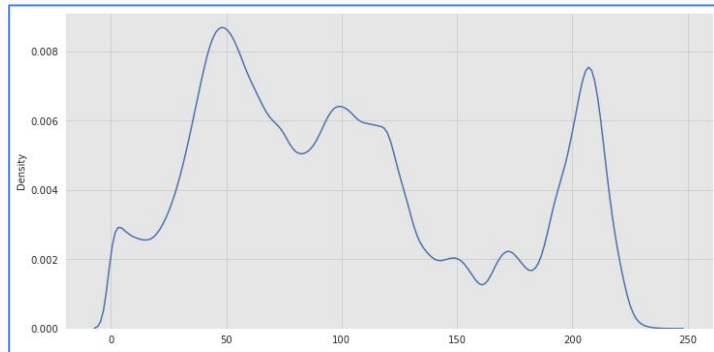
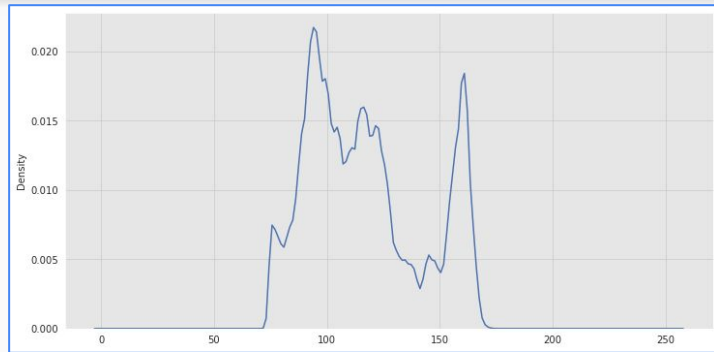
More Example



- Histograms of a color image
- The distribution of the three color channels are calculated separately

Contrast Stretching

Histogram Point of View



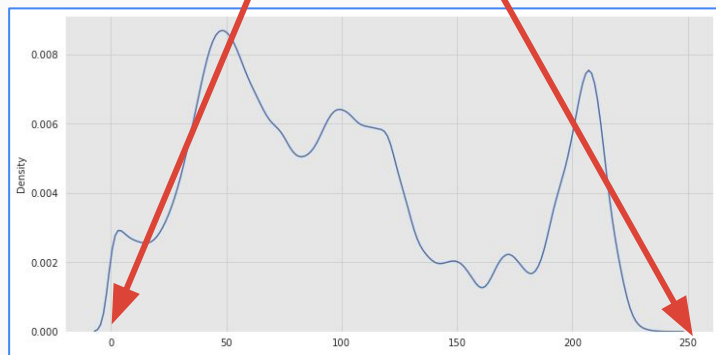
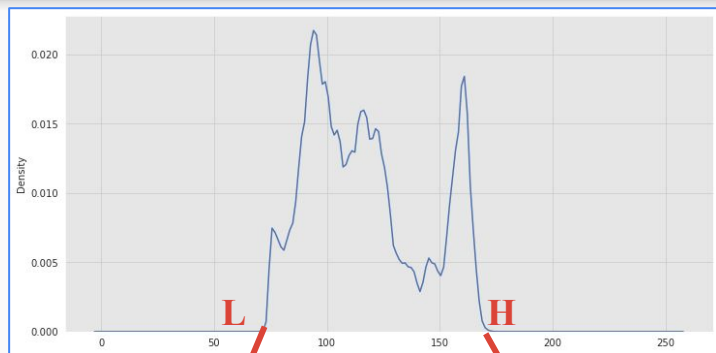
$s = T(r)?$

Contrast Stretching

Histogram Point of View

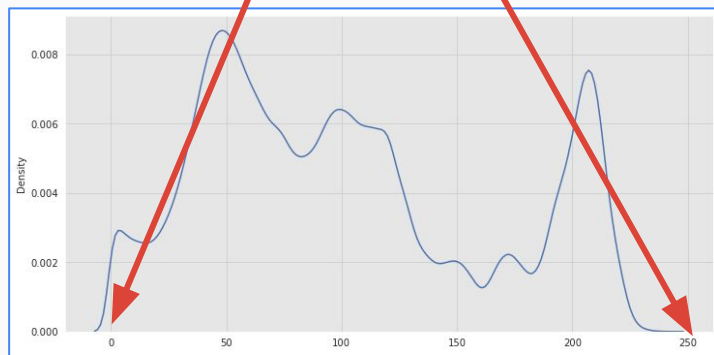
$L \rightarrow 0$ $H \rightarrow 255$

$s = T(r)?$



Contrast Stretching

Histogram Point of View



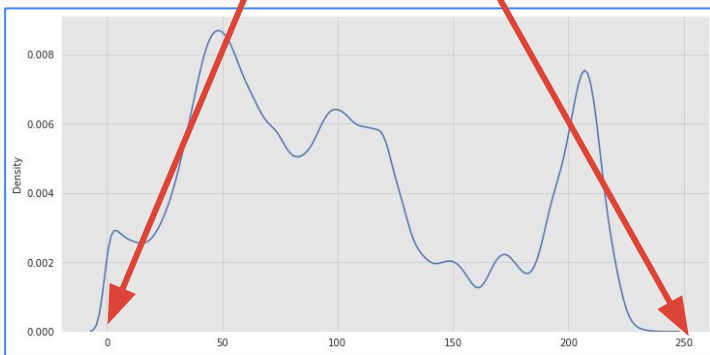
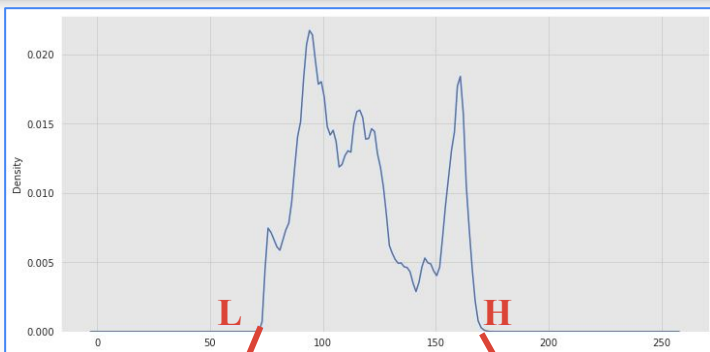
$s = T(r)?$

$$L \rightarrow 0 \quad H \rightarrow 255$$

$$s = (r - L) \times \frac{255 - 0}{H - L}$$

Contrast Stretching

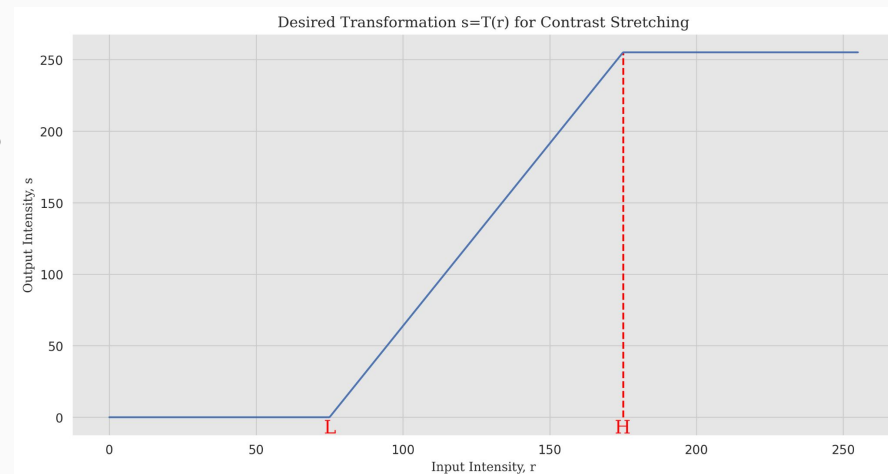
Histogram Point of View



$s = T(r)?$

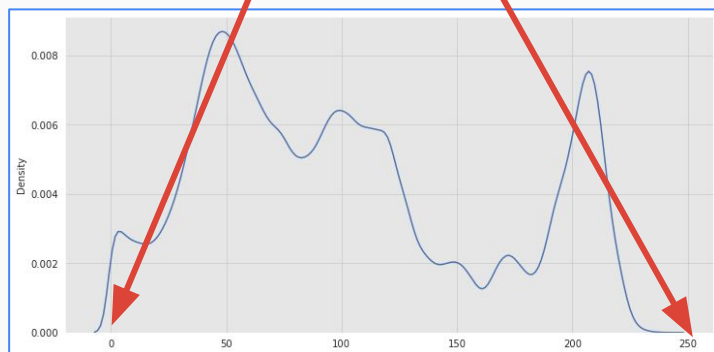
$$L \rightarrow 0 \quad H \rightarrow 255$$

$$s = (r - L) \times \frac{255 - 0}{H - L}$$

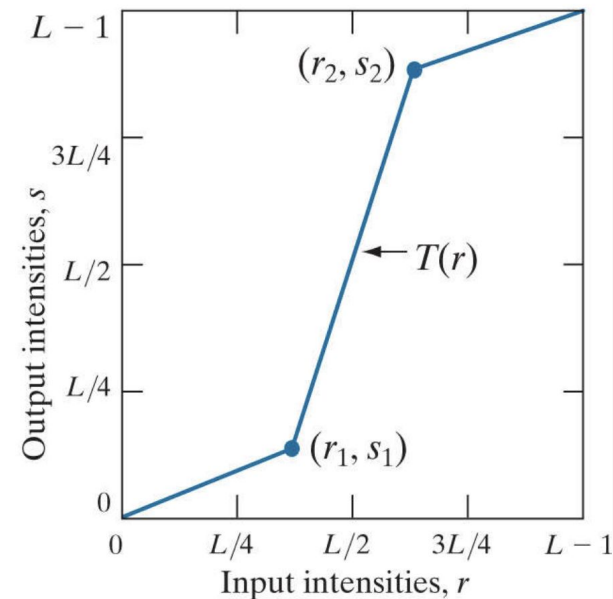


Contrast Stretching

Histogram Point of View

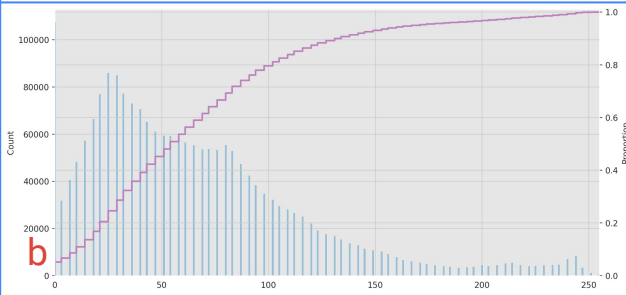
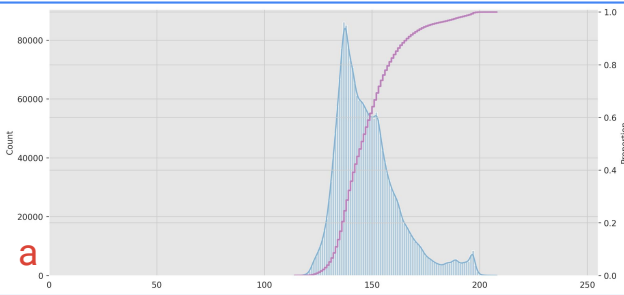


$$s = T(r)?$$



Contrast Stretching

Histogram Point of View

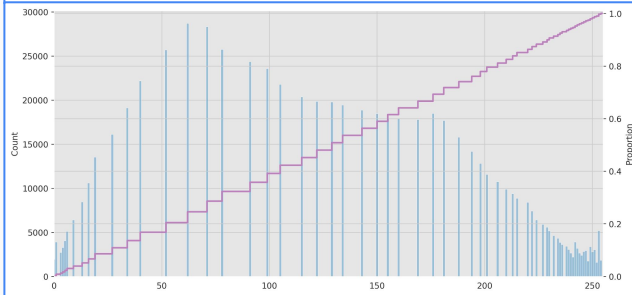
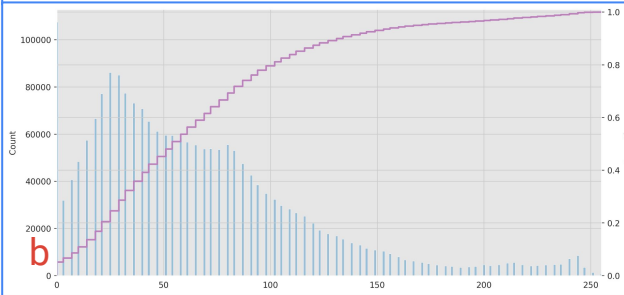
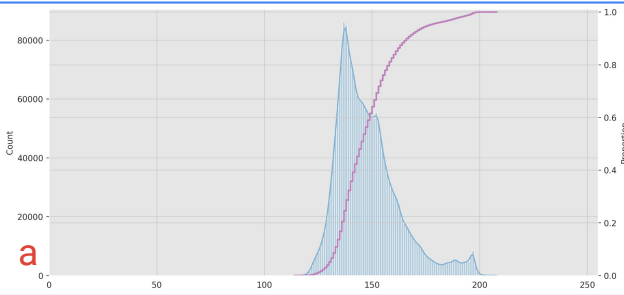


Can we do
better?

Can we do
better? - YES!

Histogram Equalization

Demo

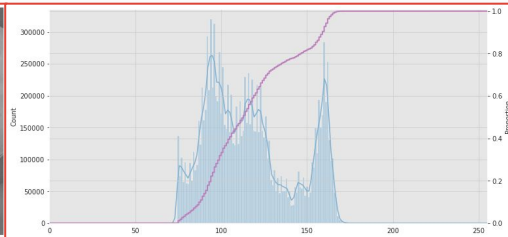


Histogram Equalization

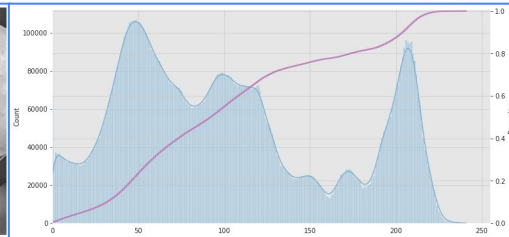
Motivation - Entropy

$$\text{Entropy}(I) = \sum_{r_k} -f(r_k) \log f(r_k)$$

- Entropy specifies the uncertainty of image pixels
- Uncertain events have more information than certain events
- Entropy is a measure of dispersion of histograms
- **Uniform distribution** - dispersed, high entropy, high contrast (desirable)



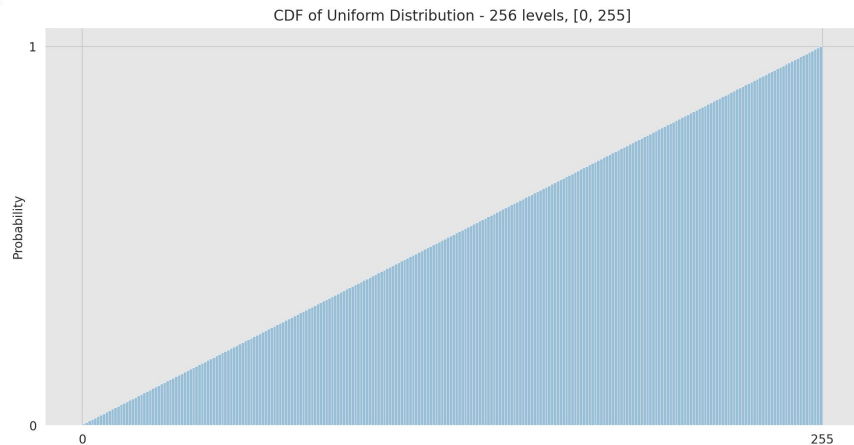
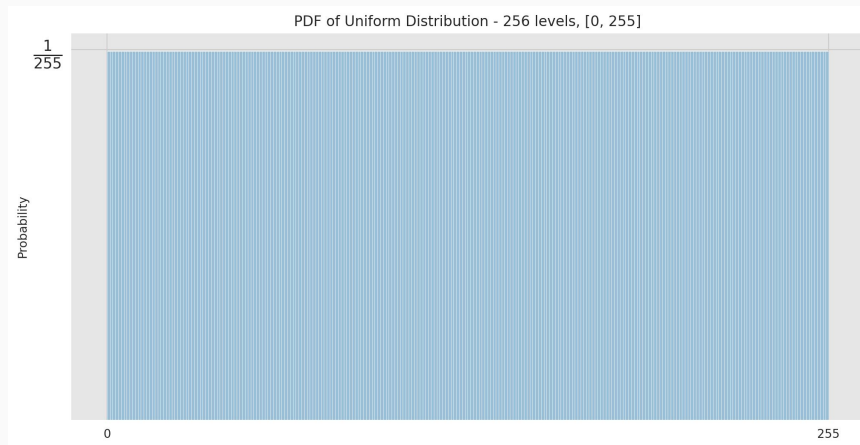
Entropy = 6.34



Entropy = 7.64

Histogram Equalization

Motivation - Uniform Distribution

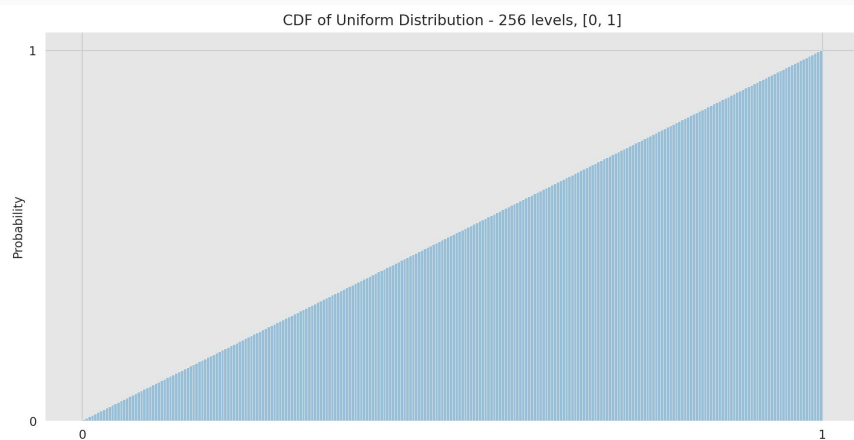
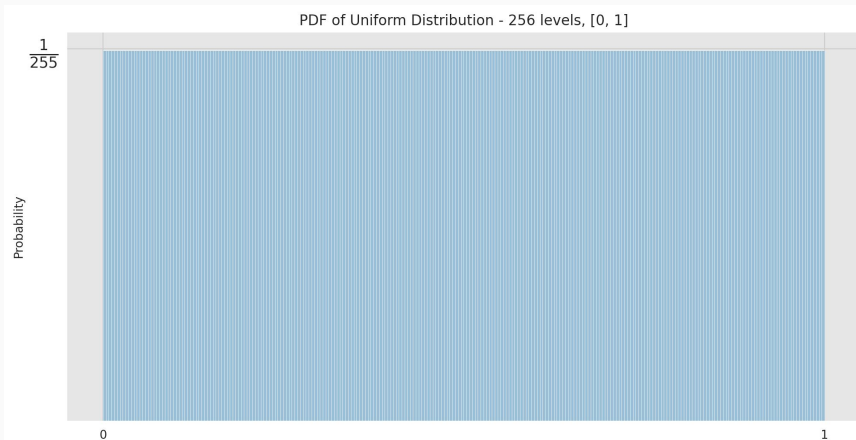


PDF is flat, $f(r_k) = \frac{1}{L-1}$, where L is the quantization level

CDF is linear between [min, max], slope = $\frac{1}{L-1}$

Histogram Equalization

Motivation - Uniform Distribution

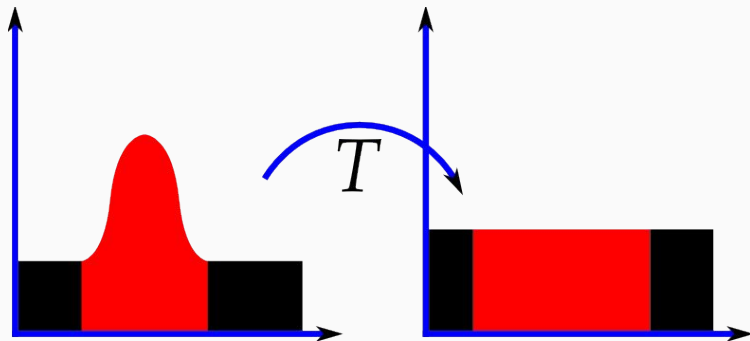


PDF is flat, $f(r_k) = \frac{1}{L-1}$, where L is the quantization level

CDF is linear between [min, max], slope = 1

Histogram Equalization

Basic Idea for Histogram Equalization

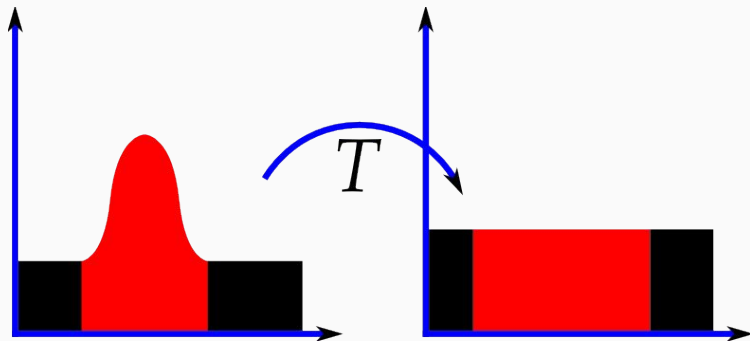


By Zefram - Own work, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=668605>

Goal: Find a map $s = T(r)$ such that the resulting PDF is flat, i.e., the resulting CDF is linear

Histogram Equalization

Basic Idea for Histogram Equalization



By Zefram - Own work, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=668605>

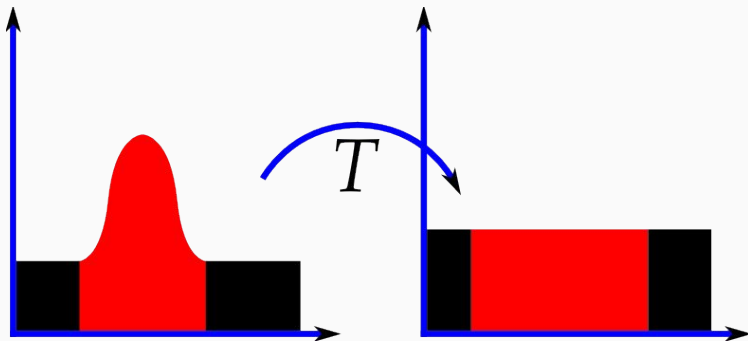
Observation (Assuming normalized gray values)

- For an ideal uniform distribution, $y\%$ pixels have values less than y [see the CDF]
- **Reverse:** A pixel with $y\%$ pixels having smaller values than itself should have a value y

Goal: Find a map $s = T(r)$ such that the resulting PDF is flat, i.e., the resulting CDF is linear

Histogram Equalization

Basic Idea for Histogram Equalization



By Zefram - Own work, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=668605>

Goal: Find a map $s = T(r)$ such that the resulting PDF is flat, i.e., the resulting CDF is linear

Observation (Assuming normalized gray values)

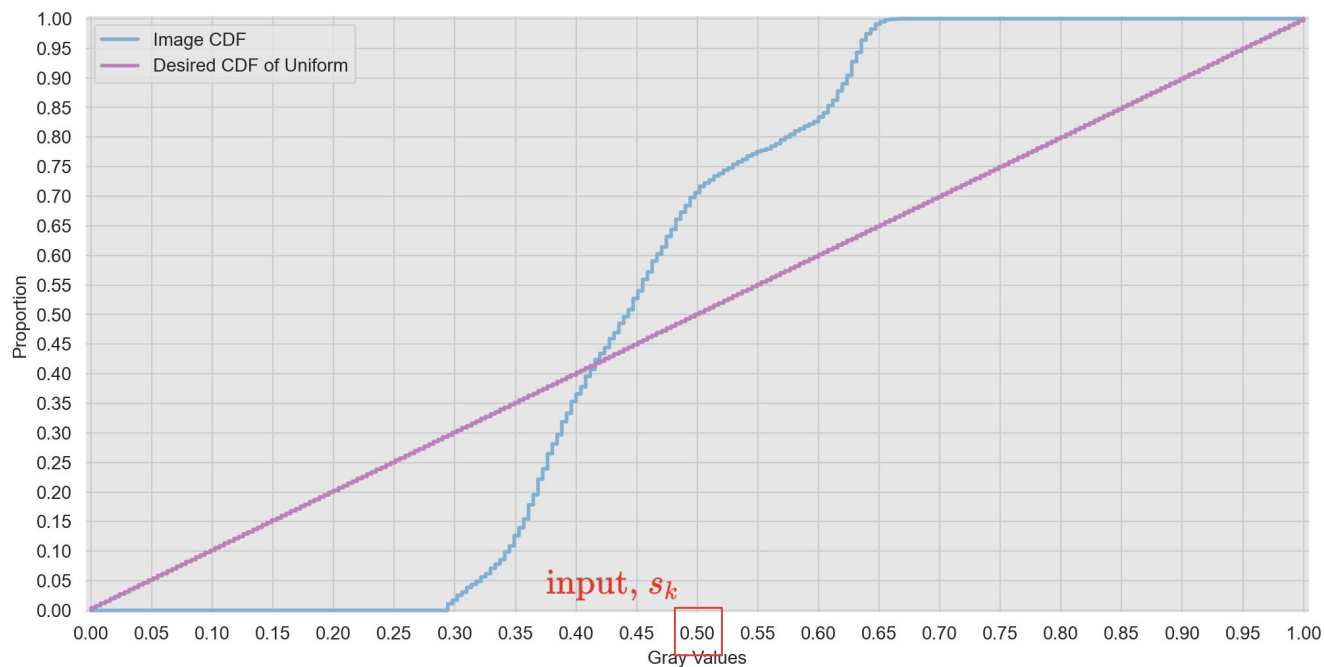
- For an ideal uniform distribution, $y\%$ pixels have values less than y [see the CDF]
- **Reverse:** A pixel with $y\%$ pixels having smaller values than itself should have a value y

Idea (Assuming normalized gray values $[0, 1]$)

- For a given gray value x , we know the percentage of pixels with values $\leq x$ (CDF)
- Hence, map x to its CDF value to get a flat histogram/linear CDF.

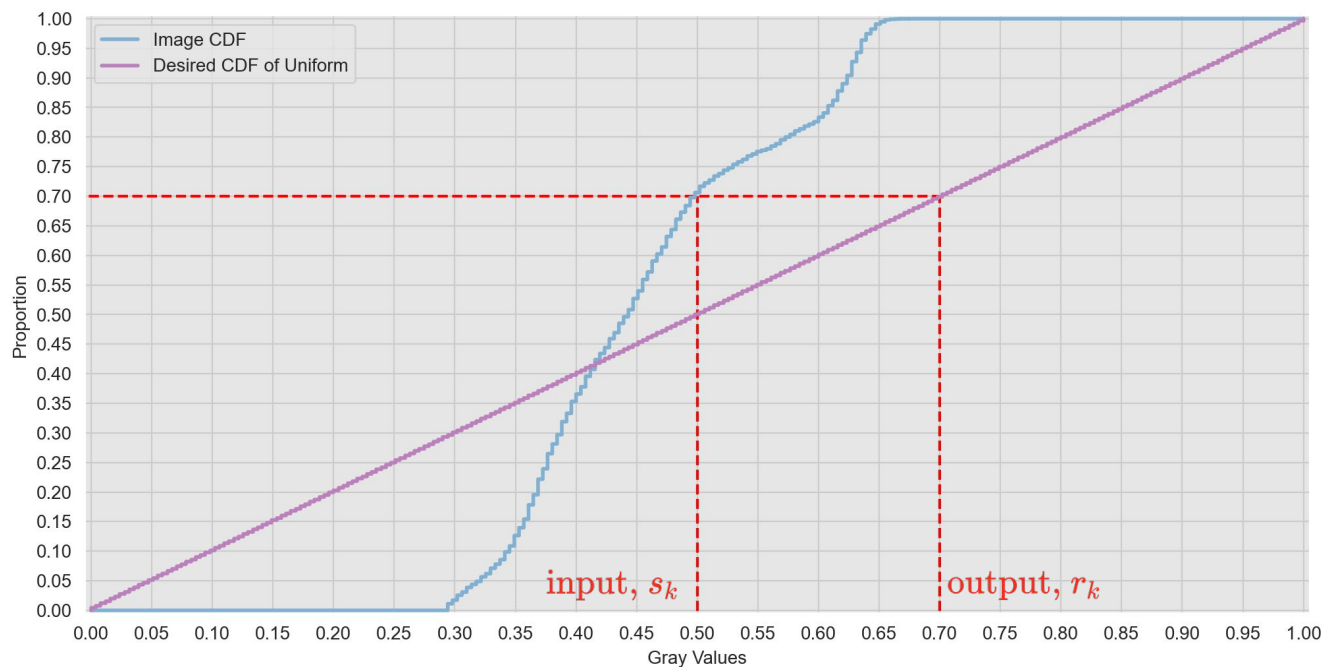
Histogram Equalization

Basic Idea for Histogram Equalization



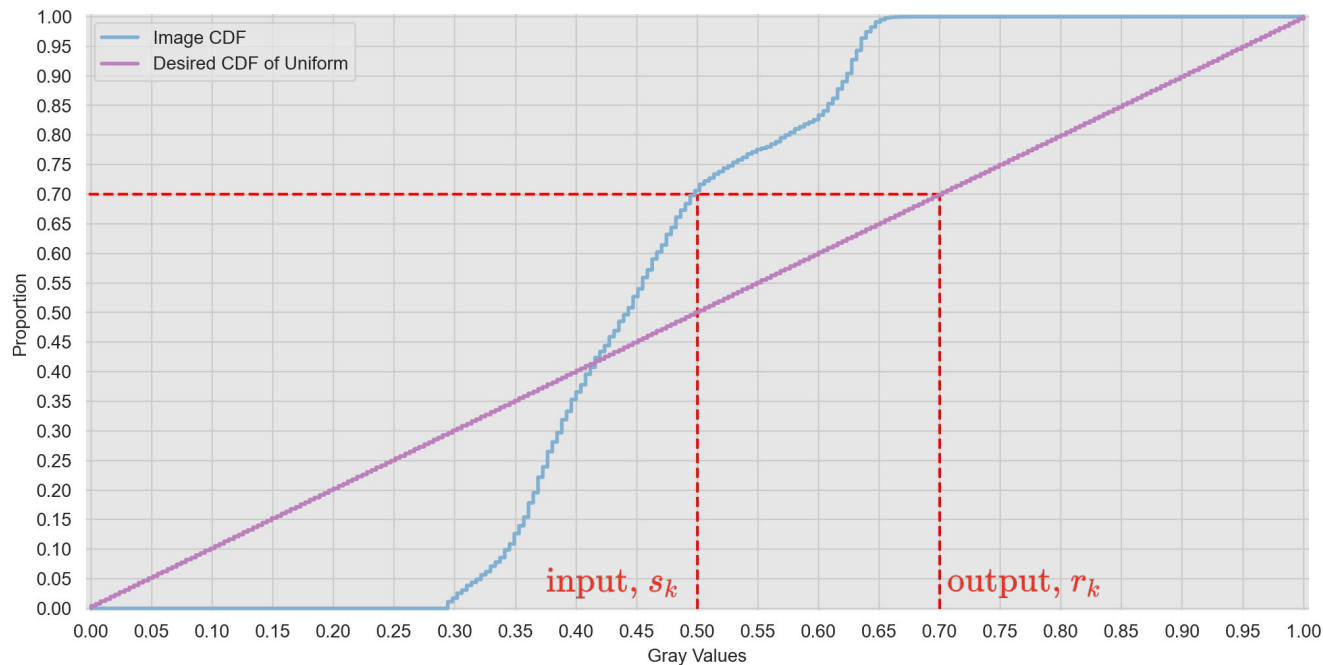
Histogram Equalization

Basic Idea for Histogram Equalization



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Hence,

$$s_k = T(r_k) = \text{CDF}_r(r_k) = F_r(r_k)$$

For L quantization levels,

$$s_k = T(r_k) = \text{round}((L - 1)F_r(r_k))$$

Histogram Equalization

Numerical Example

A random (64 x 64) image with 3 bit or 8 quantization levels ($L = 8$)

r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]
0	790	0.19	790	0.19
1	1023	0.25	1813	0.44
2	850	0.21	2663	0.65
3	656	0.16	3319	0.81
4	329	0.08	3648	0.89
5	245	0.06	3893	0.95
6	122	0.03	4015	0.98
7	81	0.02	4096	1.00

Histogram Equalization

Numerical Example

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r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	
1	1023	0.25	1813	0.44	
2	850	0.21	2663	0.65	
3	656	0.16	3319	0.81	
4	329	0.08	3648	0.89	
5	245	0.06	3893	0.95	
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r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	$= \text{round}(7 \times 0.19)$
1	1023	0.25	1813	0.44	
2	850	0.21	2663	0.65	
3	656	0.16	3319	0.81	
4	329	0.08	3648	0.89	
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r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	$=\text{round}(1.33)$
1	1023	0.25	1813	0.44	
2	850	0.21	2663	0.65	
3	656	0.16	3319	0.81	
4	329	0.08	3648	0.89	
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Numerical Example

A random (64 x 64) image with 3 bit or 8 quantization levels (**L = 8**)

r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	1
1	1023	0.25	1813	0.44	
2	850	0.21	2663	0.65	
3	656	0.16	3319	0.81	
4	329	0.08	3648	0.89	
5	245	0.06	3893	0.95	
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Histogram Equalization

Numerical Example

A random (64 x 64) image with 3 bit or 8 quantization levels ($L = 8$)

r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	1
1	1023	0.25	1813	0.44	3
2	850	0.21	2663	0.65	5
3	656	0.16	3319	0.81	6
4	329	0.08	3648	0.89	6
5	245	0.06	3893	0.95	7
6	122	0.03	4015	0.98	7
7	81	0.02	4096	1.00	7

Histogram Equalization

Numerical Example

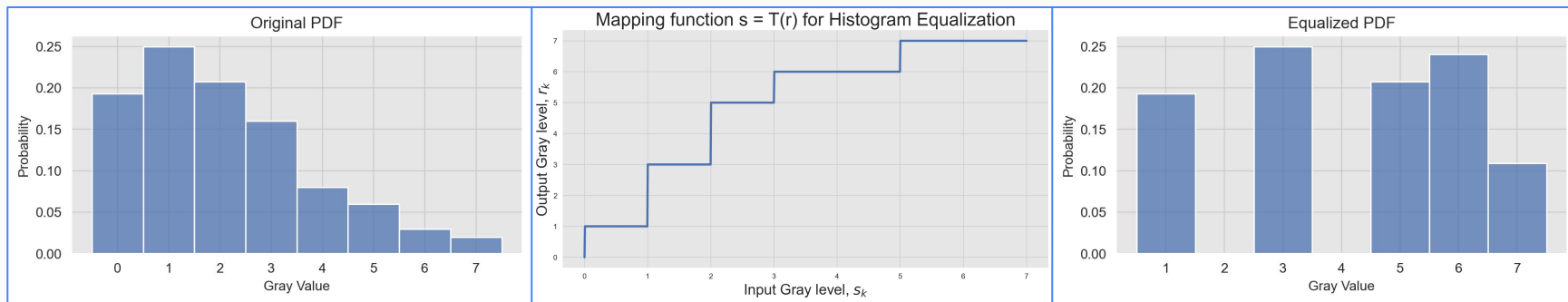
A random (64 x 64) image with 3 bit or 8 quantization levels ($L = 8$)

r_k	$h(r_k)$	$f(r_k)$ [PDF]	$H(r_k)$	$F(r_k)$ [CDF]	$s_k = \text{round}(7 \times F(r_k))$
0	790	0.19	790	0.19	1
1	1023	0.25	1813	0.44	3
2	850	0.21	2663	0.65	5
3	656	0.16	3319	0.81	6
4	329	0.08	3648	0.89	6
5	245	0.06	3893	0.95	7
6	122	0.03	4015	0.98	7
7	81	0.02	4096	1.00	7

Histogram Equalization

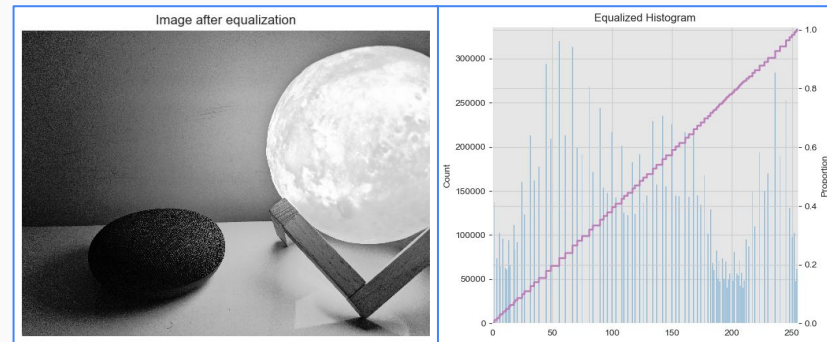
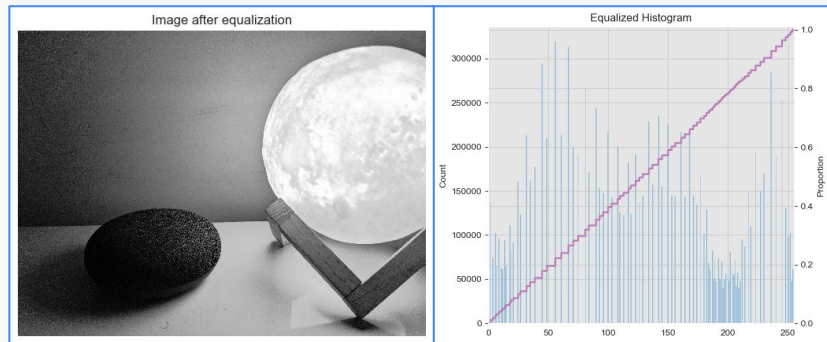
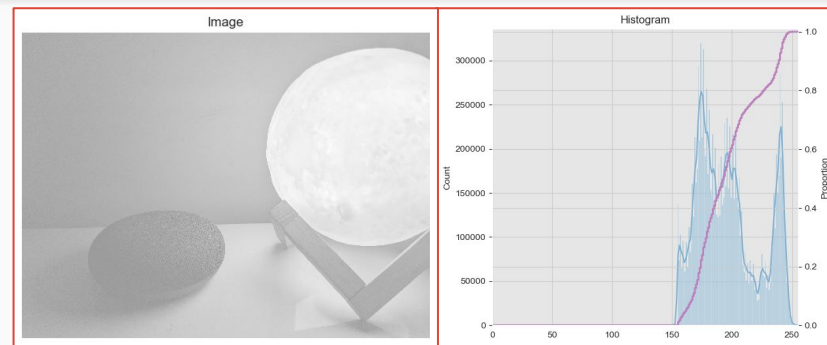
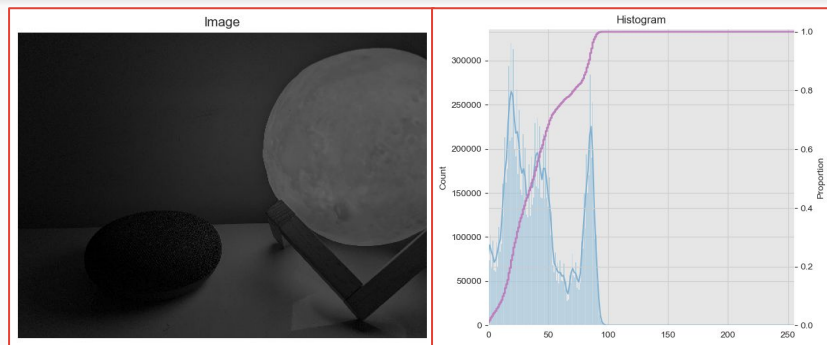
Numerical Example

A random (64 x 64) image with 3 bit or 8 quantization levels ($L = 8$)



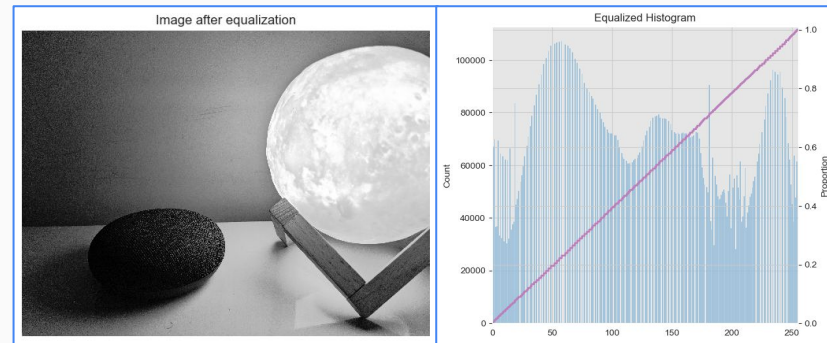
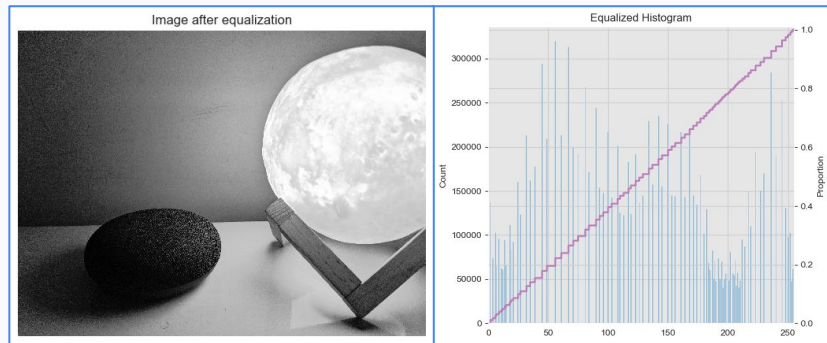
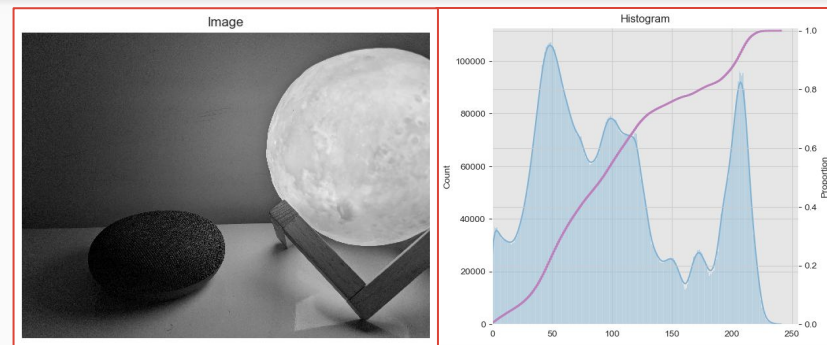
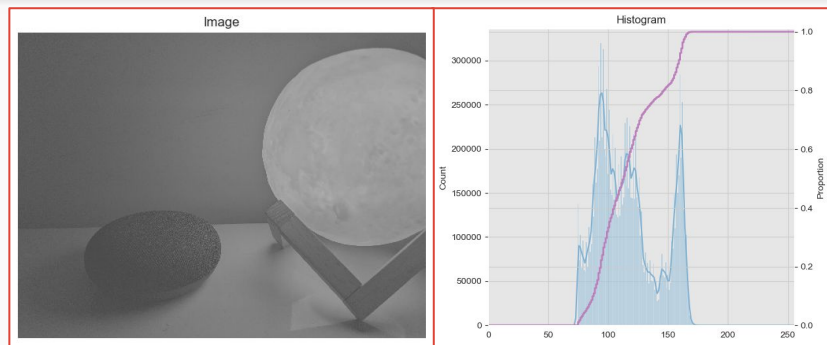
Histogram Equalization

Examples



Histogram Equalization

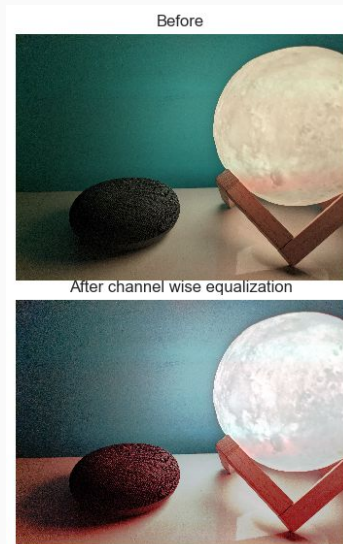
Examples



Histogram Equalization

Example

Wrong method - channel wise equalization



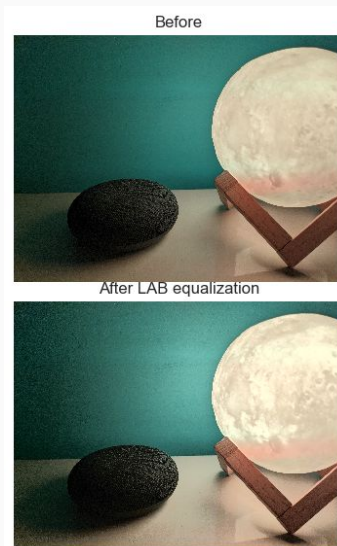
Histogram Equalization

Example

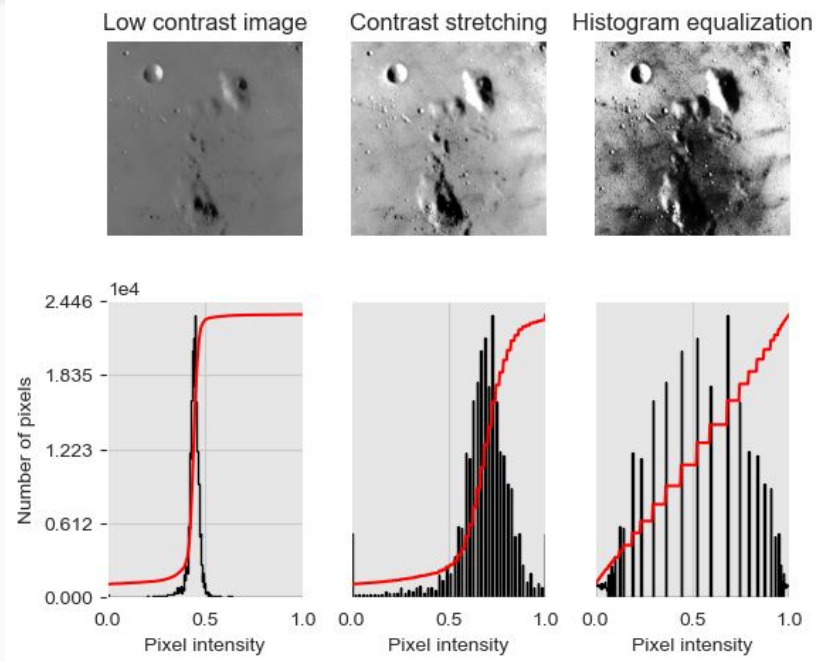
Wrong method - channel wise equalization



Correct method - RGB to LAB, equalize L, LAB to RGB

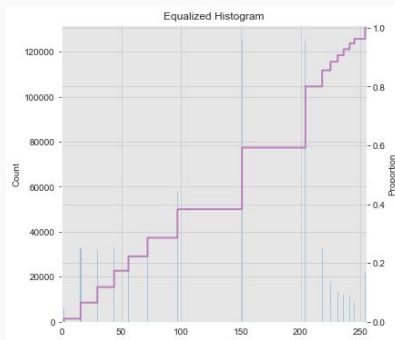
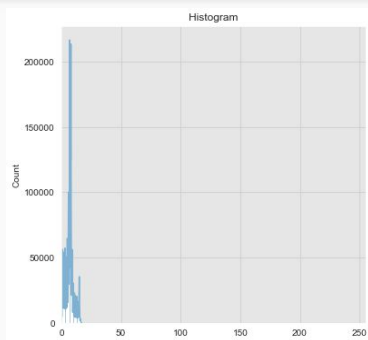
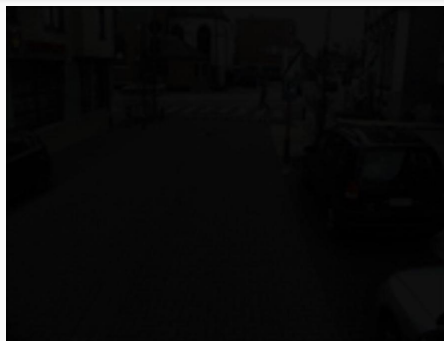


Failure of Histogram Equalization



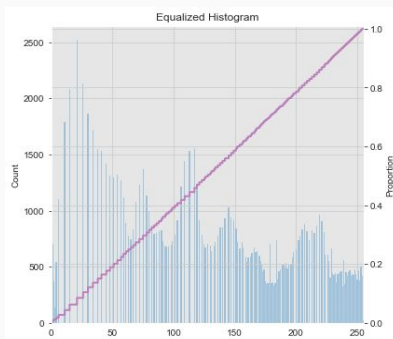
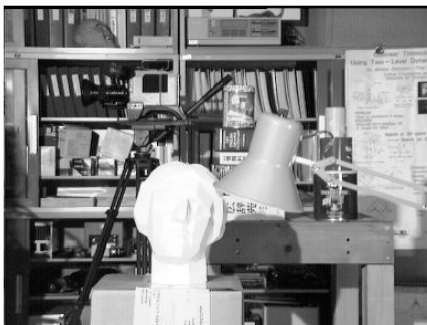
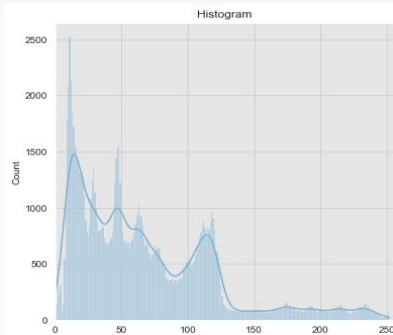
- Details in low (or high) intensity value lost
- Noise (grains) amplified

Failure of Histogram Equalization



- Does not perform well for images with large area in low intensity region
- “Equalized” histogram does not have linear CDF

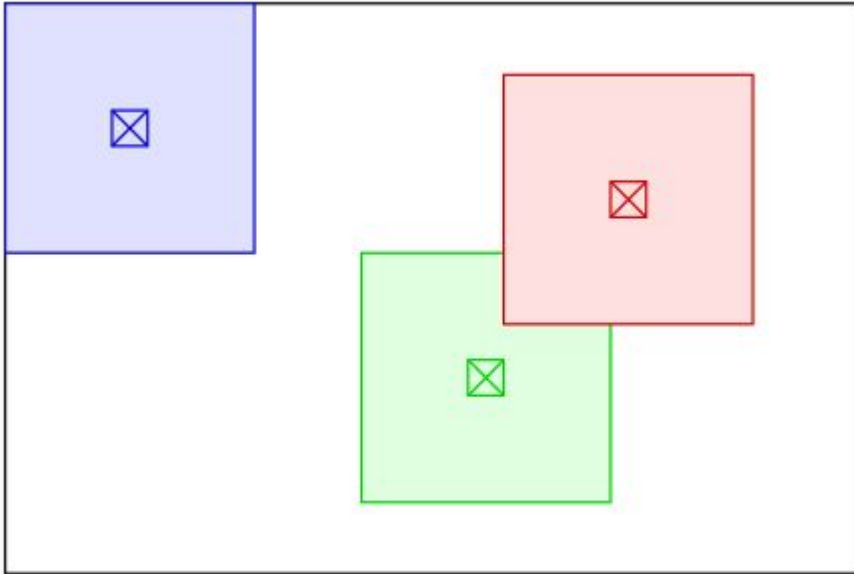
Failure of Histogram Equalization



- Flat histogram (linear CDF) not always good
- Notice - Background much clear, but face details lost

Adaptive Histogram Equalization (AHE)

Idea

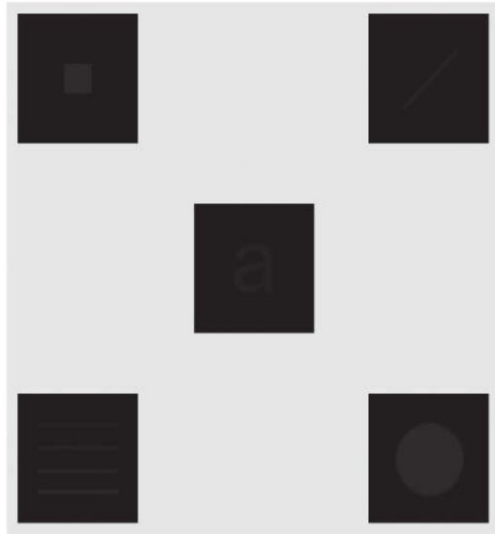


- If image contains regions that are significantly lighter (or darker), the contrast in those regions will not be sufficiently enhanced.
- Apply histogram equalization on a local neighbor

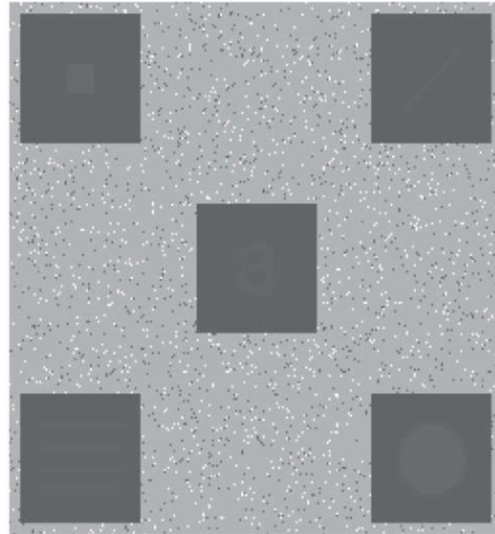
Adaptive Histogram Equalization (AHE)

Examples

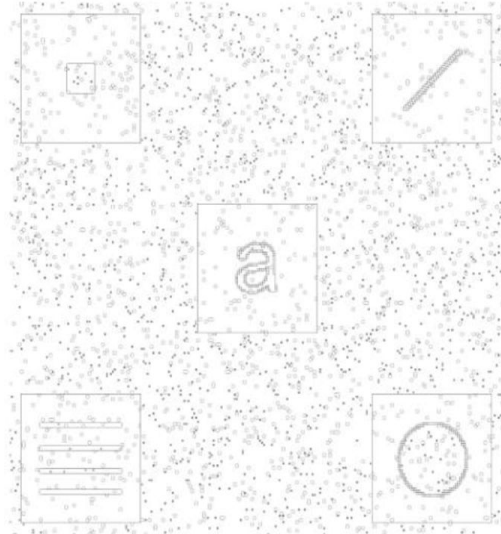
By Rafael C. Gonzalez & Richard E. Woods, 2018, *Digital Image Processing*, 4th Edition



Original Image



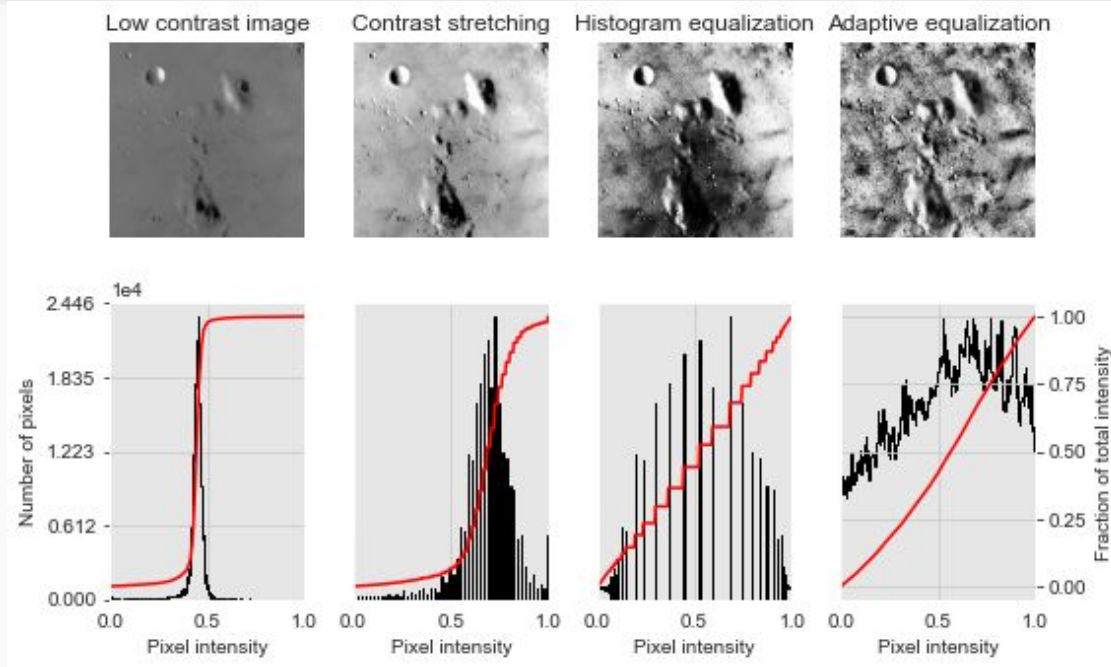
Global Histogram Equalization



Local (3x3) Histogram Equalization

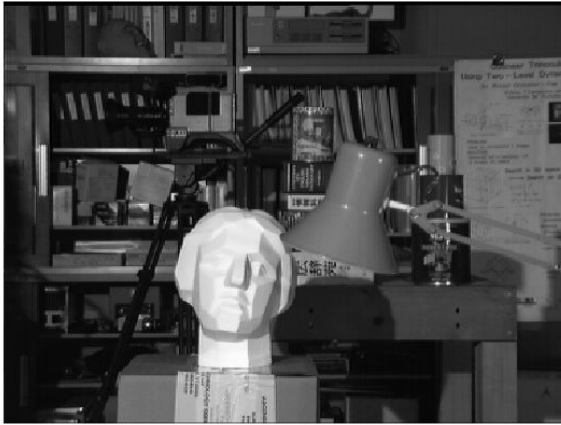
Adaptive Histogram Equalization (AHE)

Examples

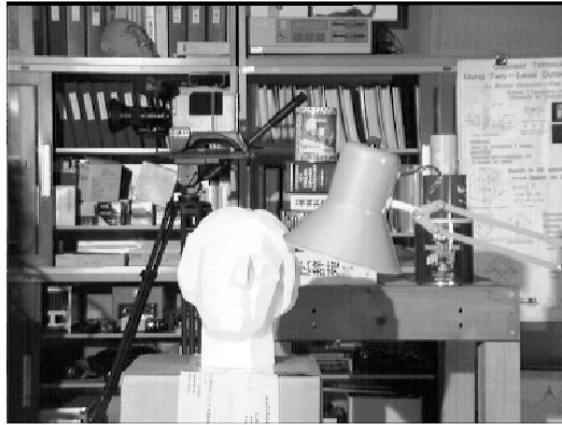


Adaptive Histogram Equalization (AHE)

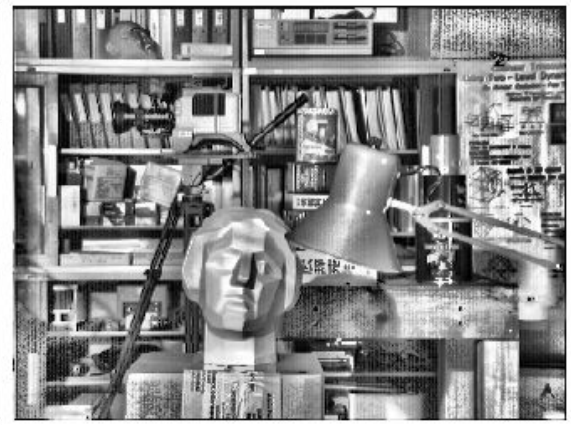
Examples



Original Image



Global Histogram Equalization



Local (36x36) Histogram Equalization

Adaptive Histogram Equalization (AHE)

Failure Example



Original Image



Global Histogram Equalization



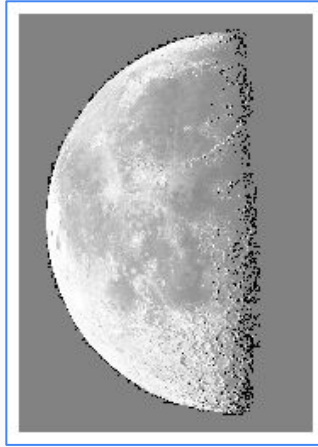
Local (100x100) Histogram Equalization

Adaptive Histogram Equalization (AHE)

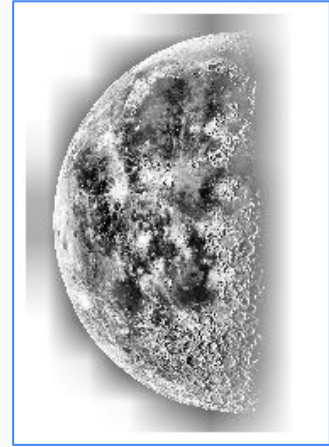
Failure Example



Original Image



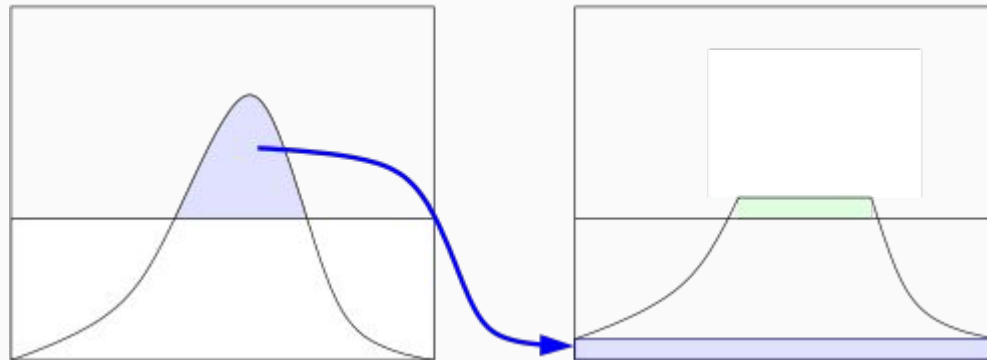
Global Histogram Equalization



Local (100x100) Histogram Equalization

Contrast Limited Adaptive Histogram Equalization (CLAHE)

- Both global and AHE tends to amplify the contrast in near-constant region
- Result - noise amplification
- Solution: Contrast Limited AHE (CLAHE), where the contrast amplification is limited



Contrast Limited Adaptive Histogram Equalization (CLAHE)



Original Image



Global Histogram Equalization



Local (100x100) Histogram Equalization

Clip limit = 1% of kernel size

Contrast Limited Adaptive Histogram Equalization (CLAHE)



Original Image



Global Histogram Equalization



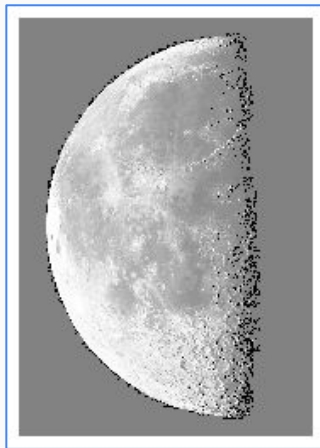
Local (100x100) Histogram Equalization

Clip limit = 1% of kernel size

Contrast Limited Adaptive Histogram Equalization (CLAHE)



Original Image



Global Histogram Equalization



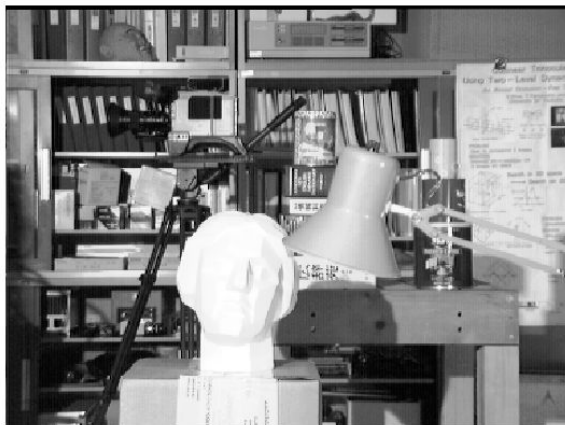
Local (100x100) Histogram Equalization

Clip limit = 5% of kernel size

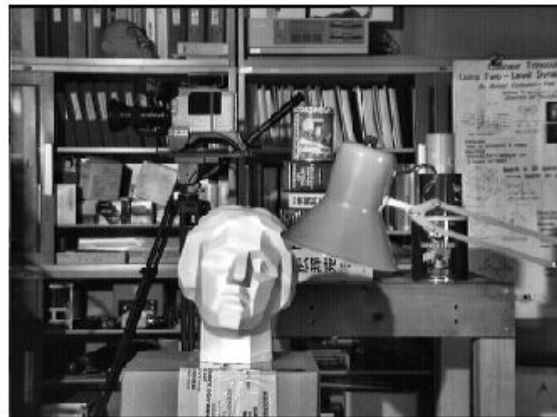
Contrast Limited Adaptive Histogram Equalization (CLAHE)



Original Image



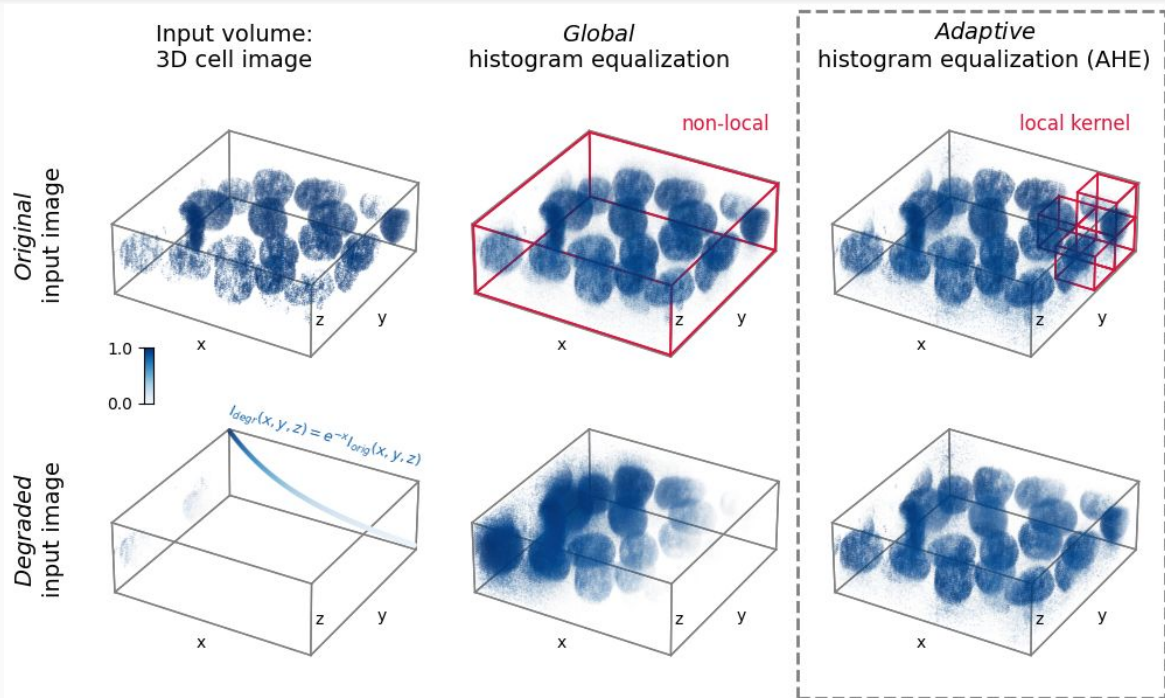
Global Histogram Equalization



Local (36x36) Histogram Equalization

Clip limit = 1% of kernel size

Contrast Limited Adaptive Histogram Equalization (CLAHE) - 3D Image



Histogram Matching (Specification)

Example

Source



Reference

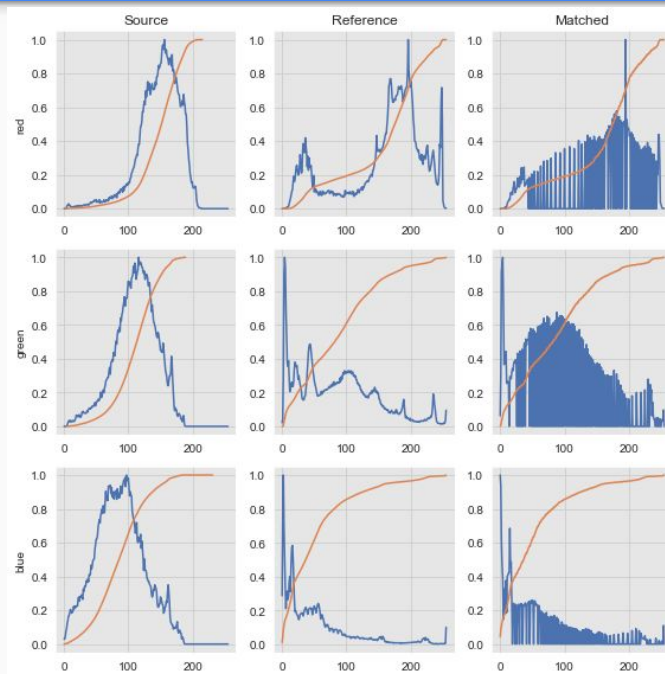


Matched



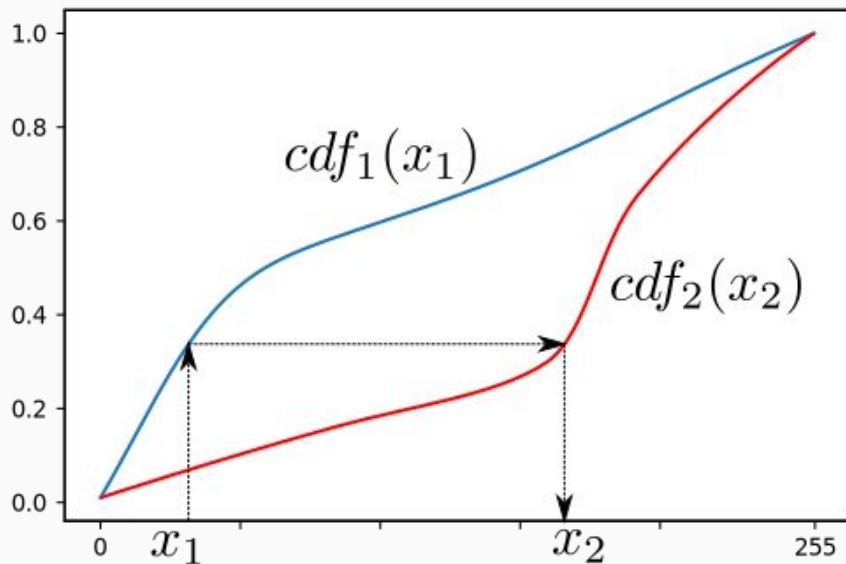
Histogram Matching (Specification)

Example



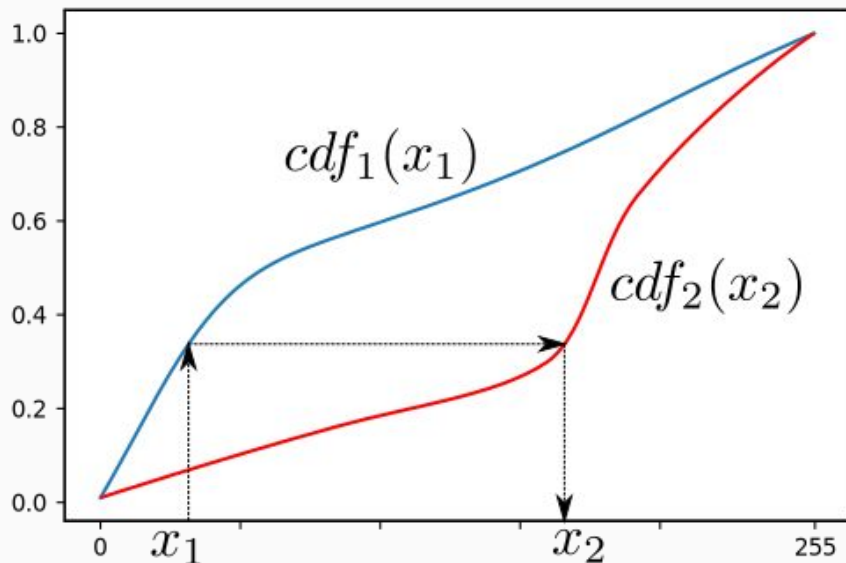
Histogram Matching

How to



Histogram Matching

How to



$$x_2 = T(x_1) = cdf_2^{-1}(cdf_1(x_1))$$

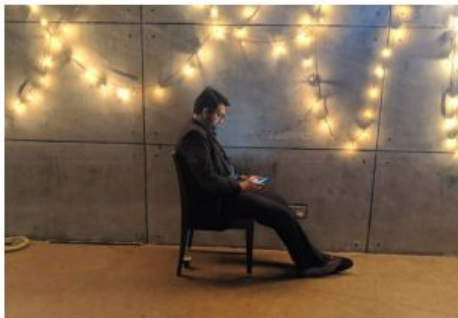
Histogram Matching

Example

Source



Reference



Matched



Thank You!